

CS 171: Problem Set 2

Due Date: February 8th, 2024 at 8:59pm via Gradescope

1. Negligible/Non-negligible functions

Let $f, g : \mathbb{N} \rightarrow [0, 1]$ be negligible functions, let $p : \mathbb{N} \rightarrow \mathbb{R}$ be a polynomial such that $p(n) > 0$ for all $n \in \mathbb{N}$.

- (a) Define $h : \mathbb{N} \rightarrow \mathbb{R}$ as $h(n) = f(n) + g(n)$. Prove that h is a negligible function.
- (b) Define $h : \mathbb{N} \rightarrow \mathbb{R}$ as $h(n) = f(n) \cdot p(n)$. Prove that h is a negligible function.

For each function below, either prove that it is negligible or prove that it is non-negligible (all logarithms are base 2).

- (c) $f(n) = n^{-100} + 2^{-n}$
- (d) $f(n) = 1.01^{-n}$
- (e) $f(n) = 2^{-(\log n)^2}$
- (f) $f(n) = e^{-\log^3 n} + e^{-\log^2 n} + e^{-\log n}$

Solution

- (a) For a given polynomial P , we want to show that there exists N_P such that for all $n > N_P$, $h(n) < \frac{1}{P(n)}$. As f and g are negligible, then we can find N_f and N_g such that $f(n) < \frac{1}{2P(n)}$ and $g(n) < \frac{1}{2P(n)}$. Thus choose $N_P = \max(N_f, N_g)$, so that for all $n > N_P$, $h(n) = f(n) + g(n) < \frac{1}{P(n)}$. Thus h is negligible.
- (b) For a given polynomial P , we want to show that there exists N_P such that for all $n > N_P$, $h(n) < \frac{1}{P(n)}$. As f is negligible, then we can find N_f such that for all $n > N_f$, $f(n) < \frac{1}{p(n)P(n)} \implies f(n) \cdot p(n) < \frac{1}{P(n)}$. Thus choose $N_P = N_f$, so that for all $n > N_P$, $h(n) = f(n) \cdot p(n) < \frac{1}{P(n)}$. Thus h is negligible.
- (c) It is non-negligible. For every n ,

$$f(n) = n^{-100} + 2^{-n} > n^{-100} = \frac{1}{n^{100}}.$$

Thus, for the polynomial $P(n) = n^{100}$, $f(n)$ is never smaller than $1/P(n)$. Therefore, f is non-negligible.

- (d) It is negligible. Let P be any positive polynomial of degree d . There is some N_0 such that

$$P(n) < n^{d+1}$$

for every $n > N_0$. We use $d+1$ simply because one extra power of n eventually dominates the coefficients and lower-degree terms of P .

Since $\log(1.01) > 0$ is a constant and $n/\log n$ grows without bound, there is some N_1 such that, for every $n > N_1$,

$$\frac{n}{\log n} \log(1.01) > d + 1.$$

Multiplying both sides by $\log n$ gives

$$n \log(1.01) > (d + 1) \log n.$$

Exponentiating both sides gives

$$1.01^n > n^{d+1} > P(n).$$

Therefore, for every $n > \max\{N_0, N_1\}$,

$$f(n) = 1.01^{-n} < \frac{1}{P(n)},$$

so f is negligible.

- (e) It is negligible. Again, let P be any positive polynomial of degree d . For all sufficiently large n , we have both $P(n) < n^{d+1}$ and $\log n > d + 1$. It follows that

$$2^{(\log n)^2} > 2^{(d+1) \log n} = n^{d+1} > P(n).$$

Taking reciprocals gives

$$f(n) = 2^{-(\log n)^2} < \frac{1}{P(n)}.$$

Since this holds for every positive polynomial P , f is negligible.

- (f) It is non-negligible. All three terms are positive, so

$$f(n) \geq e^{-\log n}.$$

Recall that all logarithms in the problem are base 2. Since $e < 4$ and $\log_2 n > 0$ for $n > 1$, we have

$$f(n) \geq e^{-\log_2 n} > 4^{-\log_2 n} = 2^{-2 \log_2 n} = \frac{1}{n^2}.$$

Thus $f(n)$ never becomes smaller than the inverse of the polynomial n^2 , so f is non-negligible.

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2. 2-time security?

An encryption scheme $(\text{Gen}, \text{Enc}, \text{Dec})$ over message space $\mathcal{M}$ and ciphertext space $\mathcal{C}$ is said to be 2-time perfectly secure if for any $(m_1, m_2) \in \mathcal{M} \times \mathcal{M}$ and $(m'_1, m'_2) \in \mathcal{M} \times \mathcal{M}$ such that $m_1 \neq m_2$ and $m'_1 \neq m'_2$ and for any $c_1, c_2 \in \mathcal{C}$ the following holds:

$$\Pr[\text{Enc}(K, m_1) = c_1 \wedge \text{Enc}(K, m_2) = c_2] = \Pr[\text{Enc}(K, m'_1) = c_1 \wedge \text{Enc}(K, m'_2) = c_2].$$

Note that in the above definition the key K is the same for encrypting m_1, m_2 (resp. m'_1, m'_2). Consider the following encryption scheme for the message space $\mathbb{Z}_{23}$.

- **Gen:** Sample two elements $a \xleftarrow{\$} \mathbb{Z}_{23}$ and $b \xleftarrow{\$} \mathbb{Z}_{23}$.
- **Enc** $((a, b), m)$: Output $c = a \cdot m + b \pmod{23}$.
- **Dec** $((a, b), c)$: Compute $m = (c - b) \cdot a^{-1} \pmod{23}$ if a is invertible. Otherwise, output *error*.

Show the following.

1. Prove that for any message $m \in \mathbb{Z}_{23}$,

$$\Pr[\text{Dec}(K, \text{Enc}(K, m)) = m] = \frac{22}{23}$$

2. Prove that this is 2-time secure.

Solution

1. Let us use A to denote the random variable for a in the encryption scheme above.
If $A \neq 0$, then the decryption algorithm recovers the message. If $A = 0$, then the decryption algorithm outputs *error*. $\Pr[A \neq 0] = \frac{22}{23}$.
2. (a) **Variables:** Fix any set of messages $(m_0, m_1, m'_0, m'_1) \in \mathbb{Z}_{23}$ such that $m_0 \neq m_1$ and $m'_0 \neq m'_1$. Also fix ciphertexts $c_1, c_2 \in \mathbb{Z}_{23}$. Next, let $A \leftarrow \mathbb{Z}_{23}$ and $B \leftarrow \mathbb{Z}_{23}$ be independent random variables. Let $K = (A, B)$ be the random variable denoting the key.
- (b) The following events are equivalent:

$$\text{Enc}(K, m_1) = c_1 \wedge \text{Enc}(K, m_2) = c_2 \tag{0.1}$$

$$A \cdot m_1 + B = c_1 \wedge A \cdot m_2 + B = c_2 \tag{0.2}$$

$$A = \frac{c_1 - c_2}{m_1 - m_2} \wedge B = c_1 - m_1 \cdot \frac{c_1 - c_2}{m_1 - m_2} \tag{0.3}$$

To get event 0.3, we solved the system of equations for A and B . The system of equations has a unique solution because $m_1 - m_2 \neq 0$.

- (c) In event 0.3, A and B are set equal to deterministic values. Since A and B are sampled independently:

$$\begin{aligned} \Pr \left[A = \frac{c_1 - c_2}{m_1 - m_2} \wedge B = c_1 - m_1 \cdot \frac{c_1 - c_2}{m_1 - m_2} \right] &= \Pr \left[A = \frac{c_1 - c_2}{m_1 - m_2} \right] \cdot \Pr \left[B = c_1 - m_1 \cdot \frac{c_1 - c_2}{m_1 - m_2} \right] \\ &= \frac{1}{23} \cdot \frac{1}{23} = \frac{1}{23^2} \end{aligned}$$

(d) By the same argument, we can show that

$$\Pr[\text{Enc}(K, m'_1) = c_1 \wedge \text{Enc}(K, m'_2) = c_2] = \frac{1}{2 \cdot 3^2}$$

Therefore,

$$\Pr[\text{Enc}(K, m_1) = c_1 \wedge \text{Enc}(K, m_2) = c_2] = \Pr[\text{Enc}(K, m'_1) = c_1 \wedge \text{Enc}(K, m'_2) = c_2]$$

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3. Pseudorandom Functions

Let $f : \{0, 1\}^n \times \{0, 1\}^n \rightarrow \{0, 1\}^n$ be a pseudorandom function (PRF). For the functions f' below, either prove that f' is a PRF (for all choices of f), or prove that f' is not a PRF.

(a) $f'_k(x) := f_k(0||x)||f_k(1||x)$.

(b) $f'_k(x) := f_k(0||x)||f_k(x||1)$.

Solution Recall that the PRF experiment chooses a random bit: in one case the adversary receives oracle access to the keyed function, and in the other it receives oracle access to a uniformly random function with the same domain and range. The experiment outputs 1 exactly when the adversary guesses the chosen bit correctly. Security requires this probability to be at most $1/2 + \text{negl}(n)$ for every PPT adversary. Let $F = \{f_k\}_k$ denote the original keyed family and let $F' = \{f'_k\}_k$ denote the family being analyzed. We write the experiment using the course notation $\text{PRF}_{\mathcal{A}, F}(n)$.

(a) Yes, f' is a PRF. Suppose there is a PPT adversary $\mathcal{A}$ whose advantage in the PRF experiment for F' is some non-negligible function $\varepsilon(n)$. In other words,

$$\Pr[\text{PRF}_{\mathcal{A}, F'}(n) = 1] \geq \frac{1}{2} + \varepsilon(n).$$

We construct a PPT adversary $\mathcal{B}$ for the PRF experiment for F .

The adversary $\mathcal{B}$ runs $\mathcal{A}$. Whenever $\mathcal{A}$ queries $x \in \{0, 1\}^{n-1}$, $\mathcal{B}$ queries its own oracle O at $0||x$ and $1||x$, and returns

$$O(0||x)||O(1||x)$$

to $\mathcal{A}$. Finally, $\mathcal{B}$ outputs the same guess as $\mathcal{A}$.

If $O = f_k$, then $\mathcal{A}$ receives oracle access to f'_k . If O is a uniformly random function from $\{0, 1\}^n$ to $\{0, 1\}^n$, then the inputs $0||x$ and $1||x$ used for distinct values of x are all distinct. Consequently, the pairs $O(0||x)||O(1||x)$ are independent and uniformly random elements of $\{0, 1\}^{2n}$. Thus, in this case, $\mathcal{A}$ receives oracle access to a uniformly random function from $\{0, 1\}^{n-1}$ to $\{0, 1\}^{2n}$.

Let b be the hidden bit in the experiment. We use the convention that $b = 0$ selects the keyed function and $b = 1$ selects a uniformly random function. Recall that $\mathcal{B}$ simply outputs whatever $\mathcal{A}$ outputs. We consider the two possible values of b separately.

If $b = 0$, then $\mathcal{B}$'s oracle is f_k . Therefore, every query x made by $\mathcal{A}$ is answered with

$$f_k(0||x)||f_k(1||x) = f'_k(x).$$

This is exactly the view that $\mathcal{A}$ receives when its hidden bit is 0. Therefore,

$$\Pr[\mathcal{B} \text{ outputs } 0 \mid b = 0] = \Pr[\mathcal{A} \text{ outputs } 0 \mid b = 0].$$

If $b = 1$, then $\mathcal{B}$'s oracle is a uniformly random function R . As argued above,

$$R'(x) = R(0||x)||R(1||x)$$

is a uniformly random function from $\{0, 1\}^{n-1}$ to $\{0, 1\}^{2n}$. This is exactly the view that $\mathcal{A}$ receives when its hidden bit is 1. Therefore,

$$\Pr[\mathcal{B} \text{ outputs } 1 \mid b = 1] = \Pr[\mathcal{A} \text{ outputs } 1 \mid b = 1].$$

The experiment $\text{PRF}_{\mathcal{B}, F}(n)$ outputs 1 when $\mathcal{B}$ guesses its hidden bit correctly. Since the hidden bit is uniform, we can now expand this probability using the two cases:

$$\begin{aligned} \Pr[\text{PRF}_{\mathcal{B}, F}(n) = 1] &= \frac{1}{2} \Pr[\mathcal{B} \text{ outputs } 0 \mid b = 0] + \frac{1}{2} \Pr[\mathcal{B} \text{ outputs } 1 \mid b = 1] \\ &= \frac{1}{2} \Pr[\mathcal{A} \text{ outputs } 0 \mid b = 0] + \frac{1}{2} \Pr[\mathcal{A} \text{ outputs } 1 \mid b = 1] \\ &= \Pr[\text{PRF}_{\mathcal{A}, F'}(n) = 1] \\ &\geq \frac{1}{2} + \varepsilon(n). \end{aligned}$$

Since ε is non-negligible, this contradicts the security of f , which requires $\mathcal{B}$'s success probability to be at most $1/2 + \text{negl}(n)$. Hence f' is a PRF.

(b) No, f' is not a PRF. Define

$$x_0 = 0^{n-1} \quad \text{and} \quad x_1 = 0^{n-2}1.$$

The adversary queries its oracle O at these two points and writes

$$O(x_0) = y_0||y_1 \quad \text{and} \quad O(x_1) = z_0||z_1,$$

where each block has length n . It guesses that the oracle is f'_k if $y_1 = z_0$, and guesses that the oracle is random otherwise.

To see why this test works, suppose first that $O = f'_k$. The two oracle responses are

$$\begin{aligned} f'_k(x_0) &= f_k(0^n)||f_k(0^{n-1}1), \\ f'_k(x_1) &= f_k(0^{n-1}1)||f_k(0^{n-2}11). \end{aligned}$$

The second half of the first response is therefore always equal to the first half of the second response: $y_1 = z_0$.

Now suppose that O is a uniformly random function. Since $x_0 \neq x_1$, the strings $O(x_0)$ and $O(x_1)$ are independent uniform $2n$ -bit strings. In particular, y_1 and z_0 are independent uniform n -bit strings, so

$$\Pr[y_1 = z_0] = 2^{-n}.$$

More explicitly, let the hidden bit be $b = 0$ when the oracle is f'_k and $b = 1$ when the oracle is random. The adversary outputs $b' = 0$ when $y_1 = z_0$, and outputs $b' = 1$ otherwise. From the two cases above,

$$\Pr[b' = b \mid b = 0] = 1 \quad \text{and} \quad \Pr[b' = b \mid b = 1] = 1 - 2^{-n}.$$

Since the challenger chooses b uniformly, the adversary's overall success probability is

$$\Pr[\text{PRF}_{\mathcal{A}, F'}(n) = 1] = \frac{1}{2} \cdot 1 + \frac{1}{2} (1 - 2^{-n}) = 1 - 2^{-(n+1)}.$$

Its advantage over random guessing is

$$\left(1 - 2^{-(n+1)}\right) - \frac{1}{2} = \frac{1}{2} - 2^{-(n+1)},$$

which is non-negligible. Thus its success probability is not bounded by $1/2 + \text{negl}(n)$, so f' is not a PRF. ■

4. Weak CPA Security

Consider a weaker definition of CPA security where in the indistinguishability experiment the adversary $\mathcal{A}$ is not given oracle access to $\text{Enc}_k(\cdot)$ after choosing m_0, m_1 . That is, $\mathcal{A}$ can only query $\text{Enc}_k(\cdot)$ in phase 1, but not in phase 2. We call this definition weak-CPA-security. Prove that weak-CPA-security is equivalent to CPA-security (i.e., Definition 3.22 in the textbook).

Hint: Begin by showing via a hybrid argument that any $\mathcal{A}$ interacting in the usual CPA game cannot distinguish whether its phase 2 queries are answered honestly (that is, if the response to the query m is $\text{Enc}_k(m)$ or an encryption of 0; $\text{Enc}_k(0)$).

Solution Recall that weak-CPA security says that every PPT adversary $\mathcal{B}$ satisfies

$$\Pr[\text{PrivK}_{\mathcal{B}, \Pi}^{\text{wcpa}}(n) = 1] \leq \frac{1}{2} + \text{negl}(n).$$

The direction from CPA security to weak-CPA security is immediate, since a weak-CPA adversary is simply a CPA adversary that makes no phase-2 queries. We prove the other direction.

Assume that Π is weak-CPA secure. To prove that it is also CPA secure, suppose, toward a contradiction, that there is a PPT adversary $\mathcal{A}$ that breaks its CPA security. Thus, for some non-negligible function ε ,

$$\Pr[\text{PrivK}_{\mathcal{A}, \Pi}^{\text{cpa}}(n) = 1] \geq \frac{1}{2} + \varepsilon(n).$$

Let $q = q(n)$ be a polynomial upper bound on the number of phase-2 queries made by $\mathcal{A}$. We want to use $\mathcal{A}$ to construct an attacker $\mathcal{B}$ that breaks weak-CPA security. However, we cannot do this directly: after $\mathcal{B}$ forwards $\mathcal{A}$'s challenge messages to its weak-CPA challenger, $\mathcal{B}$ loses access to its encryption oracle, while $\mathcal{A}$ still expects its phase-2 queries to be answered. Let's consider the following attacker:

Attacker $\mathcal{B}'$

1. Run $\mathcal{A}$ and forward all of its phase-1 encryption queries.
2. When $\mathcal{A}$ outputs (m_0, m_1) , forward these messages to the weak-CPA challenger and give the resulting challenge ciphertext to $\mathcal{A}$.
3. Forward all of $\mathcal{A}$'s phase-2 encryption queries and return the responses.
4. Output the bit output by $\mathcal{A}$.

When running $\mathcal{B}'$, $\mathcal{A}$ gets exactly its ordinary CPA experiment, so

$$\Pr[\mathcal{B}' \text{ guesses the challenge bit}] = \Pr[\text{PrivK}_{\mathcal{A}, \Pi}^{\text{cpa}}(n) = 1] \geq \frac{1}{2} + \varepsilon(n).$$

$\mathcal{B}'$ is a CPA attacker, not a weak-CPA attacker. The attacker we want is the following: fix a message $0 \in \mathcal{M}$ and define $\mathcal{B}$ by replacing every phase-2 response with a fresh encryption of 0.

Attacker $\mathcal{B}$

1. Run $\mathcal{A}$ and forward all of its phase-1 encryption queries.
2. Before requesting the challenge, obtain q fresh encryptions of 0 from the encryption oracle.
3. When $\mathcal{A}$ outputs (m_0, m_1) , forward these messages to the weak-CPA challenger and give the resulting challenge ciphertext to $\mathcal{A}$.
4. Answer each phase-2 query with one of the encryptions of 0, regardless of the queried message.
5. Output the bit output by $\mathcal{A}$.

$\mathcal{B}$ is a weak-CPA adversary because it makes all of its encryption oracle queries before submitting (m_0, m_1) to the challenger. After receiving the challenge ciphertext, it answers $\mathcal{A}$ using the encryptions of 0 that it has already obtained and makes no further oracle queries. Let its success probability be

$$p_0 = \Pr[\text{PrivK}_{\mathcal{B}, \Pi}^{\text{wcpa}}(n) = 1].$$

Since Π is weak-CPA secure, there is a negligible function μ such that

$$p_0 \leq \frac{1}{2} + \mu(n).$$

Unlike $\mathcal{B}'$, however, $\mathcal{B}$ does not give $\mathcal{A}$ its ordinary CPA view: in phase 2, $\mathcal{A}$ is not receiving the encryptions it expects. The key is that, for every $\mathcal{A}$, the views produced by $\mathcal{B}$ and $\mathcal{B}'$ are computationally indistinguishable. To prove this, we use a sequence of hybrids between $\mathcal{B}$ and $\mathcal{B}'$. For every $i \in \{0, \dots, q\}$, hybrid H_i is the following:

Hybrid H_i

1. Answer all of $\mathcal{A}$'s phase-1 queries honestly.
2. When $\mathcal{A}$ outputs (m_0, m_1) , choose $b \xleftarrow{\$} \{0, 1\}$ and return an encryption of m_b .
3. Answer the first i phase-2 queries honestly.
4. Answer the remaining $q - i$ phase-2 queries with fresh encryptions of 0.
5. Output 1 if and only if $\mathcal{A}$ outputs b .

For each $i \in \{0, \dots, q\}$, let

$$p_i = \Pr[H_i(n) = 1].$$

The endpoint H_0 is exactly the view produced by the attacker $\mathcal{B}$, so this definition of p_0 agrees with the one above. The endpoint H_q is exactly the ordinary CPA experiment described by $\mathcal{B}'$. Therefore,

$$p_0 \leq \frac{1}{2} + \mu(n) \quad \text{and} \quad p_q \geq \frac{1}{2} + \varepsilon(n).$$

It follows that $p_q - p_0 \geq \varepsilon(n) - \mu(n)$, which is non-negligible. On the other hand,

$$p_q - p_0 \leq \sum_{i=0}^{q-1} |p_{i+1} - p_i|.$$

Since q is polynomial, there must be some $i \in \{0, \dots, q-1\}$ for which

$$|p_{i+1} - p_i|$$

is non-negligible. What this means, intuitively, is that changing just the $(i+1)$ -st phase-2 response from an encryption of 0 to an honest encryption significantly changes $\mathcal{A}$'s probability of predicting b . We use this gap to construct a valid weak-CPA adversary $\mathcal{C}$.

Weak-CPA adversary $\mathcal{C}$

1. Run $\mathcal{A}$ and use the encryption oracle to answer all of $\mathcal{A}$'s phase-1 queries.
2. When $\mathcal{A}$ outputs (m_0, m_1) , choose $b \xleftarrow{\$} \{0, 1\}$, query the encryption oracle on m_b , and give the resulting ciphertext to $\mathcal{A}$.
3. Use the encryption oracle to answer the first i phase-2 queries honestly.
4. When $\mathcal{A}$ makes its $(i + 1)$ -st phase-2 query m , first obtain $q - i - 1$ fresh encryptions of 0. Then submit $(0, m)$ as the weak-CPA challenge messages and return the challenge ciphertext to $\mathcal{A}$.
5. Answer the remaining phase-2 queries with the encryptions of 0.
6. Use whether $\mathcal{A}$ guessed b correctly to guess which of $(0, m)$ was encrypted.

This is a valid weak-CPA adversary. In particular, the first i phase-2 queries are still phase-1 queries from $\mathcal{C}$'s perspective: $\mathcal{C}$ has not yet requested its own challenge. It also obtains all later encryptions of 0 before requesting that challenge.

Let t be $\mathcal{C}$'s weak-CPA challenge bit, where $t = 0$ means that the challenge encrypts 0 and $t = 1$ means that it encrypts m . If $t = 0$, then $\mathcal{A}$'s view is exactly H_i . If $t = 1$, then its view is exactly H_{i+1} .

To see the intuition, consider the extreme case in which $\mathcal{A}$ always guesses b correctly in H_i and always guesses b incorrectly in H_{i+1} . If $\mathcal{A}$ guesses b correctly, then $\mathcal{C}$ knows that $\mathcal{A}$ received the view from H_i , so $t = 0$. If $\mathcal{A}$ guesses b incorrectly, then $\mathcal{C}$ knows that $\mathcal{A}$ received the view from H_{i+1} , so $t = 1$. In this extreme case, $\mathcal{C}$ recovers t with probability 1. The same idea gives $\mathcal{C}$ a non-negligible advantage whenever the gap between p_i and p_{i+1} is non-negligible.

Assume without loss of generality that $p_{i+1} \geq p_i$; otherwise, $\mathcal{C}$ reverses the output rule that follows. The adversary $\mathcal{C}$ guesses $t = 1$ when $\mathcal{A}$ guesses b correctly, and guesses $t = 0$ otherwise. Its success probability is

$$\begin{aligned}
 \Pr[\mathcal{C} \text{ guesses } t] &= \frac{1}{2} \Pr[\mathcal{A} \text{ does not guess } b \mid t = 0] + \frac{1}{2} \Pr[\mathcal{A} \text{ guesses } b \mid t = 1] \\
 &= \frac{1}{2}(1 - p_i) + \frac{1}{2}p_{i+1} \\
 &= \frac{1}{2} + \frac{p_{i+1} - p_i}{2}.
 \end{aligned}$$

Thus, in either case,

$$\Pr[\text{PrivK}_{\mathcal{C}, \Pi}^{\text{wcpa}}(n) = 1] = \frac{1}{2} + \frac{|p_{i+1} - p_i|}{2}.$$

Because $|p_{i+1} - p_i|$ is non-negligible, $\mathcal{C}$ breaks weak-CPA security. This contradicts our assumption that Π is weak-CPA secure. Therefore no PPT adversary can break CPA security, and weak-CPA security implies CPA security. $\blacksquare$